

JNTU Hyderabad

REGULAR EXAMINATION ? 2025

Antennas and Wave Propagation ? Regular Examination, 2025

Subject Code: EC602 Semester: 6 Regulation: R20

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.
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SECTION A (Unit 1: Antenna Fundamentals)

1. Discuss the engineering applications of Antenna Fundamentals. [12 Marks]
2. Describe the working principle of Antenna Fundamentals with necessary equations. [12 Marks]
3. Derive the expression related to Antenna Fundamentals and explain each term. [12 Marks]

SECTION B (Unit 2: Wire and Aperture Antennas)

4. How is Wire and Aperture Antennas implemented? Give an algorithm / procedure and explain. [12 Marks]
5. Explain Wire and Aperture Antennas with a neat diagram/sketch and a suitable example. [12 Marks]
6. Derive the expression related to Wire and Aperture Antennas and explain each term. [12 Marks]

SECTION C (Unit 3: Antenna Arrays)

7. Explain Antenna Arrays with a neat diagram/sketch and a suitable example. [12 Marks]
8. Derive the expression related to Antenna Arrays and explain each term. [12 Marks]
9. Discuss the engineering applications of Antenna Arrays. [12 Marks]

SECTION D (Unit 4: Radio Wave Propagation)

10. Describe the working principle of Radio Wave Propagation with necessary equations. [8 Marks]
11. How is Radio Wave Propagation implemented? Give an algorithm / procedure and explain. [8 Marks]
12. Explain Radio Wave Propagation with a neat diagram/sketch and a suitable example. [12 Marks]

SECTION E (Unit 5: Microwave Communication)

13. Write short notes on Microwave Communication. [12 Marks]
14. Discuss the engineering applications of Microwave Communication. [12 Marks]
15. List the advantages and limitations of Microwave Communication. Explain any two in

detail. [12 Marks]

*** End of Question Paper ***